Sangam: A Blockchain-Coordinated Framework for Privacy-Preserving Federated Learning in Healthcare

Divyanshh Kalra
Roll No. 00414815623

Shrey Khandal
Roll No. 00314815623

Tanish Garg
Roll No. 02214815623

Dr. Mohit Mittal
Project Guide

Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning), CSE-AIML-1
Maharaja Agrasen Institute of Technology, Rohini, New Delhi, India

***Abstract*—Hospitals hold the data that diagnostic models need, but privacy law and institutional risk keep that data in silos. Federated learning (FL) trains a shared model without moving records, yet standard FL trusts a single aggregation server that can fail, cheat, or go unaudited, and it offers no defence against poisoned updates or free riders. We present Sangam, a framework that moves FL coordination onto smart contracts. A coordinator contract runs each training round and records a cryptographic commitment to every local update and every global model; a reputation contract keeps a public, tamper-evident record of each hospital's contribution. Individual updates are protected by differentially private training (DP-SGD) and pairwise-masked secure aggregation, and a validation-based contribution score drives reputation-weighted aggregation. On the Breast Cancer Wisconsin (Diagnostic) dataset split non-IID across four simulated hospitals, Sangam reaches 97.63% test accuracy, within 0.1 points of centralised-server FedAvg and 1.8 points above training alone. When one hospital mounts a sign-flipping attack, FedAvg collapses to 45.96% while Sangam holds 97.02% and cuts the attacker's aggregation weight from 28.8% to 1.9%. With a privacy budget of $\varepsilon = 4$ per hospital, accuracy stays at 96.67%. A full round with four hospitals costs 668,449 gas on an Ethereum-compatible chain.**

***Index Terms*—federated learning, blockchain, smart contracts, differential privacy, secure aggregation, Byzantine robustness, healthcare AI.**

I. Introduction

Diagnostic models improve with more data and more varied data, but the records that would help most are spread across hospitals that cannot legally or ethically pool them. In India, the Digital Personal Data Protection Act, 2023 [1] raises the cost of moving personal data, and comparable rules apply elsewhere. Each hospital is therefore left training on the patients it happens to see.

Federated learning [2] addresses this by sending the model to the data. Hospitals train locally and share only parameter updates, which a server averages into a global model. FL has been applied to tumour segmentation, imaging and electronic health records across institutions [3, 4]. Its standard form, however, places complete trust in one aggregation server. That server is a single point of failure; nothing proves to participants that it aggregated honestly; it sees every individual update, which can leak training data [5, 6]; and it has no principled way to detect a hospital that sends poisoned updates [7] or contributes nothing [8]. Finally, hospitals that invest data and compute have no transparent record of their contribution, which weakens the reason to take part.

Blockchains offer decentralised coordination, an append-only audit log and programmable incentives. Prior blockchain-FL systems [9–12] each address part of the problem, but few combine on-chain coordination with both update privacy and poisoning resistance, and fewer evaluate the combination quantitatively in a clinical setting.

This paper makes four contributions:

1. **Architecture.** Sangam, a framework in which smart contracts coordinate FL rounds, commit to every update and global model, and maintain a public reputation and reward registry, while model weights stay off chain on IPFS.
2. **Layered privacy.** A protocol that combines DP-SGD at each hospital with pairwise-masked secure aggregation, so the aggregator sees only the sum and any party that sees an individual update sees a differentially private one.
3. **Robust, accountable aggregation.** A validation-based contribution score with an asymmetric reputation rule (quick to distrust, slow to trust) that drives aggregation weights and rewards.
4. **Evaluation.** Experiments over ten random seeds on a public clinical dataset covering utility, privacy, three attacks, secure aggregation cost and gas cost, with contracts tested on a local Ethereum network.

Section II reviews related work. Section III states the system and threat model. Section IV presents the design and Section V the implementation. Section VI reports preliminary results, Section VII discusses limitations, and Section VIII concludes.

II. Related Work

A. Federated learning in healthcare

FedAvg [2] is the reference FL algorithm. Kairouz *et al.* [13] survey its open problems, including non-IID data, privacy and robustness. Sheller *et al.* [3] showed that FL across ten institutions approaches the accuracy of pooled data for brain tumour segmentation, and Rieke *et al.* [4] argue that FL is a practical route to multi-institutional digital health. These works keep the central server and its trust assumptions.

B. Privacy of updates

Updates are not anonymous: gradients can be inverted to reconstruct training images [5, 6]. Secure aggregation [14] lets a server learn only the sum of client updates by having clients add pairwise masks that cancel in the sum. Differential privacy (DP) bounds what any output reveals about one record; DP-SGD [15] achieves it by clipping per-example gradients and adding Gaussian noise, with privacy loss tracked by Rényi DP accounting [16]. Opacus [17] implements DP-SGD for PyTorch. Each technique protects data but neither decentralises trust nor addresses contribution.

TABLE I
COMPARISON WITH RELATED APPROACHES

Work	Decentr. coord.	On-chain audit	Secure agg.	DP	Poison defence	Incentive
FedAvg [2]	–	–	–	–	–	–
Bonawitz [14]	–	–	●	–	–	–
FLTrust [20]	–	–	–	–	●	–
BlockFL [9]	●	●	–	–	–	●
Kang [21]	○	●	–	–	●	●
DeepChain [10]	●	●	○	–	○	●
Lu [12]	●	●	–	●	–	–
BFLC [11]	●	●	–	–	●	○
Sangam	●	●	●	●	●	●

● supported ○ partial – not addressed

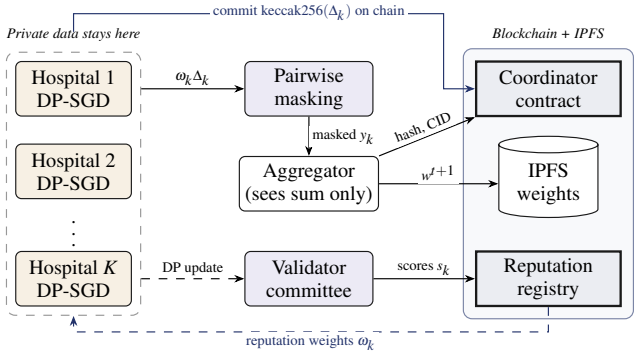


Fig. 1. Sangam architecture. Hospitals train with DP-SGD and mask their updates; the aggregator sees only the sum. Commitments, global-model hashes and reputation live on chain; weights live on IPFS.

C. Robust aggregation and incentives

Byzantine-robust rules such as Krum [18] and the coordinate-wise median [19] limit the influence of outlying updates, but they discard information under non-IID data and give no persistent record of who misbehaved. FLTrust [20] scores clients against a small server-held root dataset. Kang *et al.* [21] combine reputation with contract theory to select reliable workers. Lin *et al.* [8] characterise free-rider attacks in FL.

D. Blockchain-based federated learning

BlockFL [9] exchanges and verifies local updates through a blockchain and rewards devices through mining. DeepChain [10] provides auditability and a value-driven incentive using blockchain transactions. BFLC [11] uses a committee consensus to validate updates and limit malicious nodes. Lu *et al.* [12] pair a permissioned blockchain with FL and differential privacy for industrial IoT data sharing. Table I positions Sangam among these works.

Research gap. No single framework we found combines decentralised on-chain coordination, layered update privacy, robust contribution scoring and a transparent incentive record, evaluated together on clinical data. Sangam targets that combination.

III. System and Threat Model

A. Entities

Sangam has K *hospitals* $\mathcal{H} = \{H_1, \ldots, H_K\}$, each holding a private dataset D_k of n_k records. An *aggregator* combines masked updates and publishes global models; it can be operated by any consortium member and rotated. A small *validator committee* holds a curated validation set D_{val} and scores updates. Two smart contracts, the *coordinator* and the *reputation registry*, run on an Ethereum-compatible chain [27]. Model weights are stored on IPFS [22], addressed by content identifier (CID).

B. Adversaries

We assume (i) an *honest-but-curious aggregator* that follows the protocol but tries to learn from what it sees; (ii) up to $f < K/2$ *malicious hospitals* that may send arbitrary updates, including label-flipped, sign-flipped and scaled updates, or random noise in place of training (free riding); (iii) an *honest-but-curious validator committee*; and (iv) *public observers* who can read the whole chain. Hospitals are admitted by the consortium owner, so Sybil identities are out of scope. We do not defend against an aggregator colluding with $K-1$ hospitals to unmask the last one; secure aggregation itself cannot prevent this [14].

C. Goals

G1 Privacy: no raw record leaves a hospital, the aggregator learns only the aggregate, and anything revealed about one record is bounded by (ε, δ)-DP. **G2 Integrity:** every update and global model is bound to an on-chain commitment that cannot be altered later. **G3 Robustness:** malicious hospitals cannot substantially degrade the global model. **G4 Accountability:** each hospital's contribution and reputation are publicly recorded.

IV. Proposed Framework

Fig. 1 shows the architecture and Algorithm 1 the protocol for one round t.

A. Local training with differential privacy

Hospital k starts from the global model w^t and runs E local epochs of DP-SGD [15]. At each step it samples a batch B by including each record independently with probability $q_k = b/n_k$, clips each per-example gradient to norm C, and adds Gaussian noise:

$$\tilde{g} = \frac{1}{q_k n_k}\Big(\sum_{i \in B} \frac{g_i}{\max(1, \|g_i\|_2 / C)} + \mathcal{N}(0, \sigma_k^2 C^2 I)\Big). \quad (1)$$

The noise multiplier σ_k is calibrated with an RDP accountant [16] so that the hospital's total release over all T rounds satisfies (ε, δ)-DP. By post-processing, the update $\Delta_k^t = w_k^t - w^t$ inherits this guarantee for every party that sees it.

B. On-chain commitment

Before release, hospital k calls `submitUpdate`(t, h_k^t) with $h_k^t = \text{keccak256}(\Delta_k^t)$. The contract accepts one commitment per hospital per round, only while the round is open, and emits an event. This binds each hospital to the update that is later scored (G2).

C. Contribution scoring and reputation

The validator committee receives each DP-protected update, bounds its norm to the round median $\tau = \text{med}_j \|\Delta_j\|$, and computes two signals: agreement with the coordinate-wise

Algorithm 1 One Sangam training round t

Input: global model w^t, reputations r, registered hospitals $\mathscr{H}$

Aggregator: `openRound`(hash(w^t), CID(w^t))
for all $H_k \in \mathscr{H}$ **in parallel do**
  $\Delta_k \leftarrow$ DP-SGD(w^t, D_k, σ_k)
  `submitUpdate`(t, keccak256(Δ_k))
  send Δ_k to validator committee
Validators: $s \leftarrow$ SCORE($\{\Delta_k\}, D_{\text{val}}$); `recordScores`($t, s$)
Registry: update r by Eq. (4); publish weights ω
for all H_k with $r_k \geq \frac{1}{4}\max_j r_j$ **do**
  send masked y_k by Eq. (5)
Aggregator: $w^{t+1} \leftarrow w^t + \sum_k y_k$ (masks cancel)
Aggregator: `commitGlobalModel`(t, hash(w^{t+1}), CID); `closeRound`(t)

median update m and loss improvement on D_{val}:

$$c_k = \cos(\hat{\Delta}_k, m), \qquad \gamma_k = \frac{L(w^t) - L(w^t + \hat{\Delta}_k)}{L(w^t)}, \tag{2}$$

$$s_k = \frac{\max(0, c_k)}{\max_j \max(0, c_j)} \cdot \min\big(1, 2\,\mathrm{sig}(20\gamma_k)\big), \tag{3}$$

where $\hat{\Delta}_k$ is the norm-bounded update. A score in $[0,1]$ rewards updates that both agree with the consensus direction and improve validation loss. Reputation follows an asymmetric rule:

$$r_k \leftarrow \begin{cases} r_k/2 & \text{if } s_k < \frac{1}{2}\,\mathrm{med}_j s_j, \\ 0.8\,r_k + 0.2\,s_k & \text{otherwise,} \end{cases} \tag{4}$$

with $r_k = 0.5$ initially. One bad round halves reputation; recovery takes several good ones. Scores and reputations are written to the registry in one batched transaction, and scores accumulate as reward points (G4).

D. Reputation-weighted secure aggregation

Hospitals whose reputation falls below a quarter of the current maximum are excluded from the round. The rest receive public weights $\omega_k \propto n_k r_k$ from the registry, scale their update locally, and mask it. Each pair (i, j) derives a shared seed s_{ij} by key agreement; hospital i sends

$$y_i = \big\lfloor 2^{20}\omega_i \hat{\Delta}_i \big\rceil + \sum_{j>i} \mathrm{PRG}(s_{ij}) - \sum_{j<i} \mathrm{PRG}(s_{ji}) \pmod{2^{32}}. \tag{5}$$

The masks cancel in $\sum_i y_i$, so the aggregator recovers $\sum_i \omega_i \hat{\Delta}_i$ and nothing else (G1). It forms $w^{t+1} = w^t + \sum_i \omega_i \hat{\Delta}_i$, pins w^{t+1} to IPFS, and calls `commitGlobalModel` with its hash and CID.

E. Smart contracts

`SangamCoordinator` enforces the round life cycle (Idle, Open, Aggregated, Closed), restricts round management to the aggregator and submissions to registered hospitals, rejects duplicate and late submissions, and emits an event at every step. `ReputationRegistry` stores per-round scores, reputation in basis points, and cumulative reward points. Only 32-byte hashes, CIDs and integers are stored on chain; no model weights and no patient data.

V. Implementation and Experimental Setup

A. Prototype

Contracts are written in Solidity 0.8.24, compiled with the optimiser (200 runs) and tested with Hardhat; eight unit tests cover the full round life cycle, access control, duplicate and late submissions, and the reputation update, and all pass. The FL simulation is written in Python with NumPy; DP accounting uses Google's `dp-accounting` RDP accountant. The production design uses Flower [23] for client orchestration, Opacus [17] for DP-SGD, FastAPI for the service layer and a React dashboard, with deployment targeted at the Polygon Amoy testnet.

B. Dataset and partition

We use the Breast Cancer Wisconsin (Diagnostic) dataset [24]: 569 fine-needle-aspirate cases with 30 real-valued nuclear features, 212 malignant and 357 benign. For each seed we hold out a stratified 20% test set (114 cases) and a 7% validation set (57 cases) for the validator committee. The remaining 398 cases are split across $K = 4$ hospitals by a Dirichlet label partition with $\alpha = 0.5$ [25], producing skewed class mixes and hospital sizes; each hospital receives at least 8% of each class. Features are standardised.

C. Model and hyper-parameters

The model is a 30–16–1 multilayer perceptron (513 parameters) with ReLU and a sigmoid output. Training runs $T = 30$ rounds with $E = 2$ local epochs, expected batch size 16, learning rate 0.1 and clipping norm $C = 1$. All results are means over ten seeds with standard deviations. We report accuracy, ROC-AUC and F1 (benign as the positive class, label 1 in the dataset).

D. Baselines and attacks

Centralised trains on all 398 cases pooled. *Local only* trains each hospital alone and reports the mean. *FedAvg* [2] weights by n_k. *Median* is the coordinate-wise median [19]. For robustness, hospital 4, which holds 28.8% of the training records on average, is malicious and mounts one of three attacks: *label flipping* (trains on inverted labels), *sign flipping* (sends $-5\Delta_k$) or *free riding* (sends Gaussian noise whose norm matches the previous global update, without training).

VI. Preliminary Results

A. Utility

Table II shows that federation recovers almost all the benefit of pooling. Training alone gives 95.86% accuracy; Sangam reaches 97.63%, within 0.35 points of centralised training and 0.09 points of FedAvg with a trusted server. Sangam's lower variance (0.88 vs. 1.12) reflects reputation weighting damping erratic updates from small, skewed hospitals. Fig. 2 shows that Sangam converges at the same rate as FedAvg after about ten rounds.

B. Privacy–utility trade-off

Fig. 3 plots accuracy against the per-hospital budget ε with $\delta = 10^{-5}$. The calibrated noise multipliers average $\sigma = 13.1$, 7.0, 3.9 and 2.2 for $\varepsilon = 1, 2, 4, 8$. At $\varepsilon = 8$ accuracy matches the non-private model (97.63%); at $\varepsilon = 4$ it is 96.67%; even at $\varepsilon = 1$ it is 93.77%, though variance grows to 3.0 points. For $\varepsilon \geq 2$ every private configuration outperforms hospitals

TABLE II
TEST PERFORMANCE WITHOUT ATTACK (MEAN ± SD, 10 SEEDS)

Method	Accuracy (%)	ROC-AUC (%)	F1 (%)
Centralised (pooled data)	97.98 ± 1.11	99.75 ± 0.23	98.40 ± 0.88
Local only (mean of 4)	95.86 ± 1.00	99.40 ± 0.38	96.70 ± 0.81
FedAvg (trusted server)	97.72 ± 1.12	99.75 ± 0.24	98.19 ± 0.90
Coordinate-wise median	97.46 ± 1.14	99.71 ± 0.28	97.99 ± 0.90
Sangam (no DP)	**97.63 ± 0.88**	99.69 ± 0.31	98.13 ± 0.70
Sangam, DP $\varepsilon = 4$	96.67 ± 1.46	99.45 ± 0.32	97.37 ± 1.15

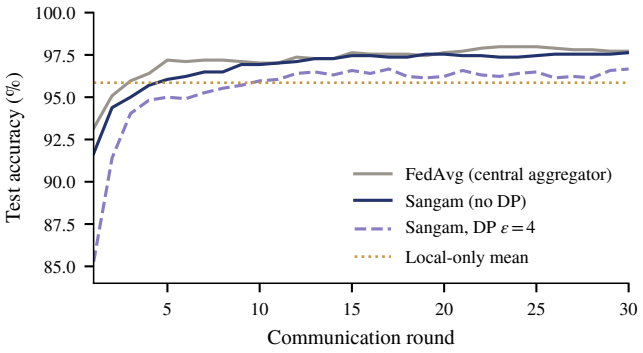


Fig. 2. Mean test accuracy per round over ten seeds.

training alone without any privacy, so a hospital gains both privacy and accuracy by joining.

C. Robustness

Fig. 4(a) shows that a single malicious hospital is enough to break FedAvg: label flipping drops it to 74.12% ± 28.07 and sign flipping to 45.96% ± 36.14, worse than a coin toss in many seeds. Sangam holds 96.93% and 97.02% respectively, on par with the coordinate-wise median (97.11% and 96.75%). Free riding does little damage to any method on this dataset.

The difference from the median is accountability. Fig. 4(b) shows the weight Sangam gives the malicious hospital. Under FedAvg it would receive 28.8% of the aggregate. Sangam reduces this to 2.4% under label flipping and 1.9% under sign flipping. The free rider's weight falls to 9.8%, about one third of its fair share: its noise earns little loss improvement but occasionally aligns with the median by chance. When the same hospital is honest it receives 29.1%, so reputation weighting does not penalise a large honest contributor. Every score behind these weights is on chain, so a hospital's exclusion can be audited and disputed.

D. Secure aggregation

For the 513-parameter model and four hospitals, the masked sum matches the plain sum to within 1.6×10^{-6}, the fixed-point quantisation error. A masked update has a correlation of 0.04 with the raw update it hides. Masking and unmasking take 0.51 ms per round versus 0.007 ms for a plain sum; the overhead grows as $O(K^2 d)$ and remains negligible beside local training.

E. On-chain cost

Table III lists measured gas on a local Hardhat network, averaged over five rounds. A full round with four hospitals costs 668,449 gas; at a gas price of 30 gwei this is about 0.020 of the chain's native token. Each hospital pays only for its

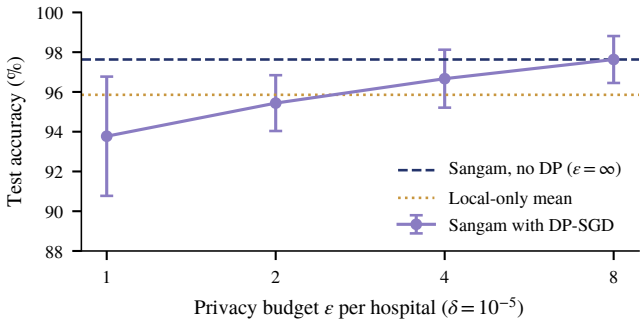


Fig. 3. Accuracy versus privacy budget (mean ± SD, 10 seeds).

TABLE III
MEASURED GAS COST (HARDHAT, OPTIMISER ON)

Operation	Gas	Called by
Deploy `SangamCoordinator`	818,749	owner, once
Deploy `ReputationRegistry`	396,462	owner, once
`registerParticipant`	47,526	owner, per hospital
`openRound`	83,117	aggregator
`submitUpdate`	58,246	each hospital
`recordScores` (4 hospitals)	197,384	validators
`commitGlobalModel`	122,571	aggregator
`closeRound`	32,393	aggregator
Total per round ($K = 4$)	**668,449**	

58,246-gas commitment. Cost is independent of model size because only hashes are stored.

VII. DISCUSSION AND LIMITATIONS

Validator visibility. Scoring requires per-hospital signals, so the validator committee sees individual updates. These are DP-protected, so leakage about any patient is bounded by ε, but the committee sees more than the aggregator. Robust secure aggregation and zero-knowledge proofs of update properties could remove this exposure.

Commitment binding. A hospital commits to the update it sends validators, but the aggregator cannot check that a masked input matches that commitment. A hospital could in principle send a different update for aggregation. Verifiable aggregation would close this gap.

Self-reported sizes. Aggregation weights use declared dataset sizes n_k, which a hospital could inflate. Reputation limits the damage, since a large but useless contribution scores poorly, but sizes are not verified.

Scale. Our evaluation uses four simulated hospitals, a small tabular dataset and a 513-parameter model. Free riding was only partly detected. Larger image tasks, more hospitals and stronger adaptive attacks remain to be tested.

VIII. CONCLUSION AND FUTURE WORK

Sangam replaces the trusted FL server with smart contracts that coordinate rounds, commit to every update and model, and keep a public contribution record, while DP-SGD and secure aggregation protect individual updates. On non-IID clinical data across four hospitals it matches trusted-server FedAvg within 0.1 points, holds 97% accuracy where FedAvg collapses to 46% under attack, and cuts a label- or sign-flipping attacker's influence by more than 90%, for under 0.7

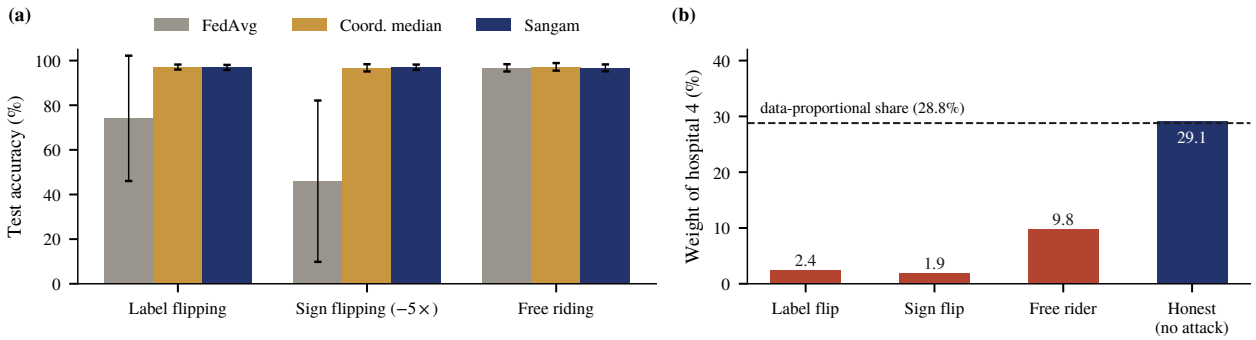


Fig. 4. Robustness with hospital 4 malicious. (a) Test accuracy under each attack (error bars: SD over 10 seeds). (b) Mean aggregation weight Sangam assigns to hospital 4 across rounds; the dashed line is its weight under plain FedAvg. The rightmost bar is the same hospital when honest.

million gas per round.

The next phase will (i) deploy the contracts to the Polygon Amoy testnet and integrate Flower and Opacus with PyTorch models; (ii) extend evaluation to medical imaging on MedMNIST [26] with more hospitals; (iii) add zero-knowledge proofs of correct aggregation; and (iv) build the monitoring dashboard and map the design to the DPDP Act and the Ayushman Bharat Digital Mission.

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